A Pre-feasibility Study for a Chili Cultivation Project

2017

JORDAN INVESTMENT COMMISSION الأردن هيئة الاستثمار

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1. Executive summary

The production of the traditional varieties of vegetables still prevails among many farmers, despite the fact that the production of non-traditional items leads to the possibility of exportation to abroad with remunerative prices benefit all stakeholders in the agricultural process. The project aims to establish a farm specialized in the production of colored chili peppers that is desired in export markets of the European Union and other European countries, the desired varaitiess will be determined in collaboration with importers in these countries, and determination of schedule to supply at the times permitted by EU marketing side. The project is based on the cultivation and production of selected varieties of chili of habanero type, and will be producing this pepper according to specifications and conditions of export of fresh vegetables and chilled. Where the project will do on drying the surplus that inalienable of export with modern ways, and then grind these products and packaged for export it and sale to shops of spices, factories, restaurants and shops.

Table 1 project summary

Project name	Production of chili
Proposed project site	Jerash
Project products	chilli Fresh and dryer too hot habanero type
Total of manpower (person)	7
Total investment (Thous.)	**106**
The net present value of cash flows (Thous.)	**170**
Payback period (years)	**4**
Internal rate of return (for 10 years)	**%26.58**

The e stimated capital costs for the project is (93,434) JD mostly of construction and processing place (24,530) JD, machinery and equipment (31817) JD, vehicles and transportation (16,500) JD and working capital (12089) JD, whereas the working capital is needed for the production requirements and workers ' salaries. Where the funding of (50%) of the project capital will be through loans and the rest through financing, despite the high investment cost of the project but its provides suitable employment opportunities suit with the scale of investment project to exploit untapped raw materials and its dependence on employment in the manufacturing process and marketing of manufactured chili.

Financial analysis results indicate that the net value of the cash flows of the project is positive, which is an indication of the financial feasibility of a project and the ratio of internal rate of return is much higher than the weighted average cost of capital, which is an indication of the feasibility of the project, and other financial indicators show that the project is financially viable and profitable for investor.

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

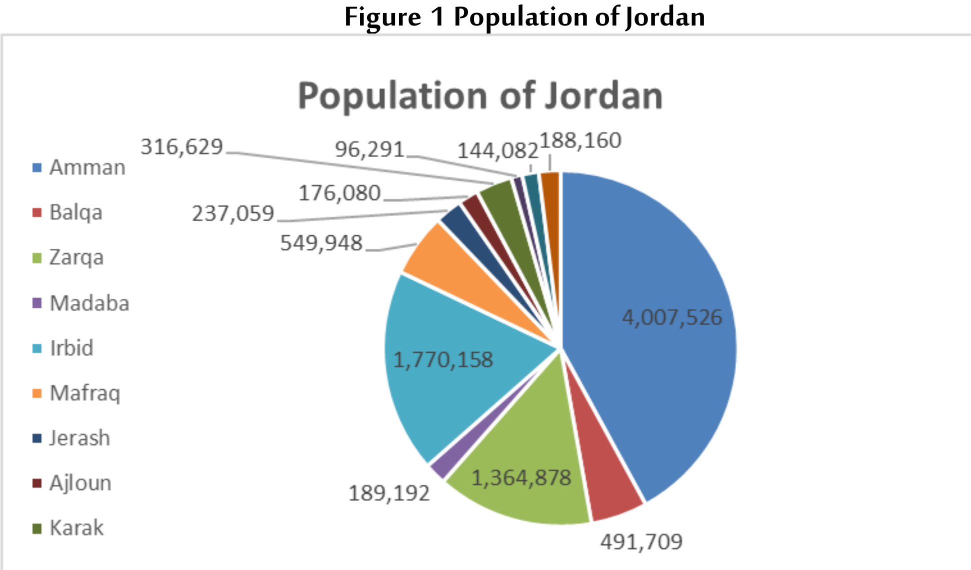
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

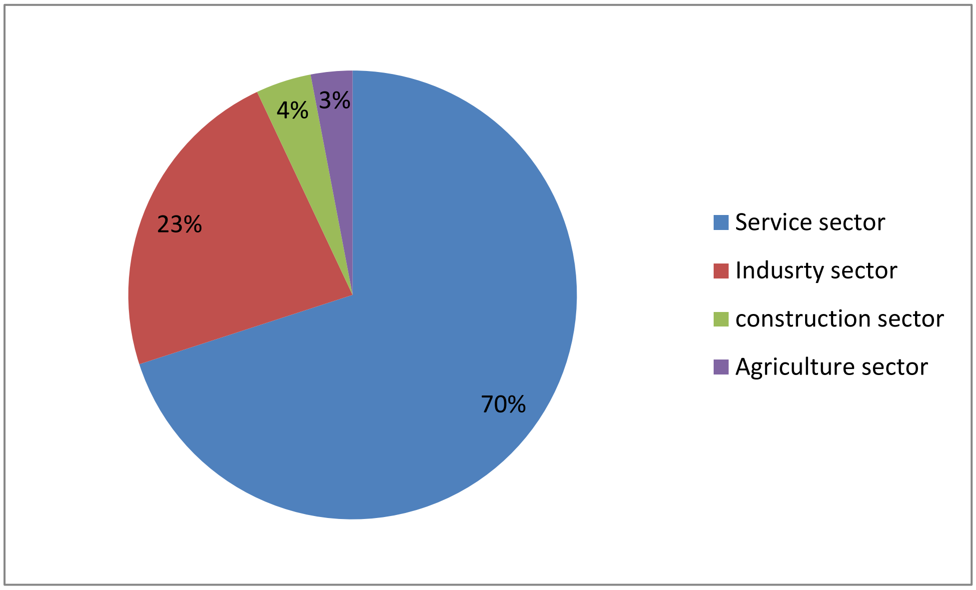
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy s lowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Jarash:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table 2 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	**Income Tax is 20%** **As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.**

2. Market analysis

Project description

Pilot projects, its based on specialization and use of technology in production in addition to understanding the internal and external market demands and try to secure the appropriate product and rely on the entry to the market in right time to achieve profits, profit achievement depends on the optimal exploitation of the climate in Jordan and the marketing support, and the availability of agricultural technical expertise.

The project based on the idea of covering the growing demand for chili in European markets by producing high quality varieties required according to the international standards, and drying the part of the non- exportable product ion , processing and packing it, and supply the domestic markets and export it in the form of dried pepper. The project aims to establish a farm for the production of varieties of colored chili, athe project consists of 10 green houses covered with plastic and protected by a special type of gauze to ensure weather conditions.
The project consists mainly of ten green houses built on an area of approximately (6) Donom of leased or owned in Jarash to produce the ripen chilies (red). And the cultivation will be renewed annually at two seasons over a ten years, the project requires to establish drying unit for pepper which non conformable to export conditions to reduce the wastage and exploit it. Where the project will work on drying the surplus of chili peppers with natural methods (solar and air) and then grind these products and packaged them for sale to the spices shops, factories and restaurants and shops.
The project is considered of agricultural service projects, it aims to diversify production and disposal of the problems and marketing bottlenecks, the project will produce the chili and supply the local market with varieties of non-traditional pepper.

Project objectives

Due to high external demand for chil's i varieties and the availability of at least 2500 regular pepper varieties, of sweet and chili peppers , and as a result of the development of consumer tastes which need increasing amounts of food and their addenda, so the project aims to produce chili at times when supply is low in European countries and produce it in the reagion between Al Shifa and Al-Aghwar in Jarash.

- Achieving good revenues for the owner.
- Production and supply of chili to the export market with distinctive quality.
- Cover part of domestic demand of dried chili.
- Excellence in production by choosing good quality and rely on modern techniques mainly is the integrated pest management, as well as the appropriate packing method and application requirements.
- Provide jobs opportunities in less developed regions.

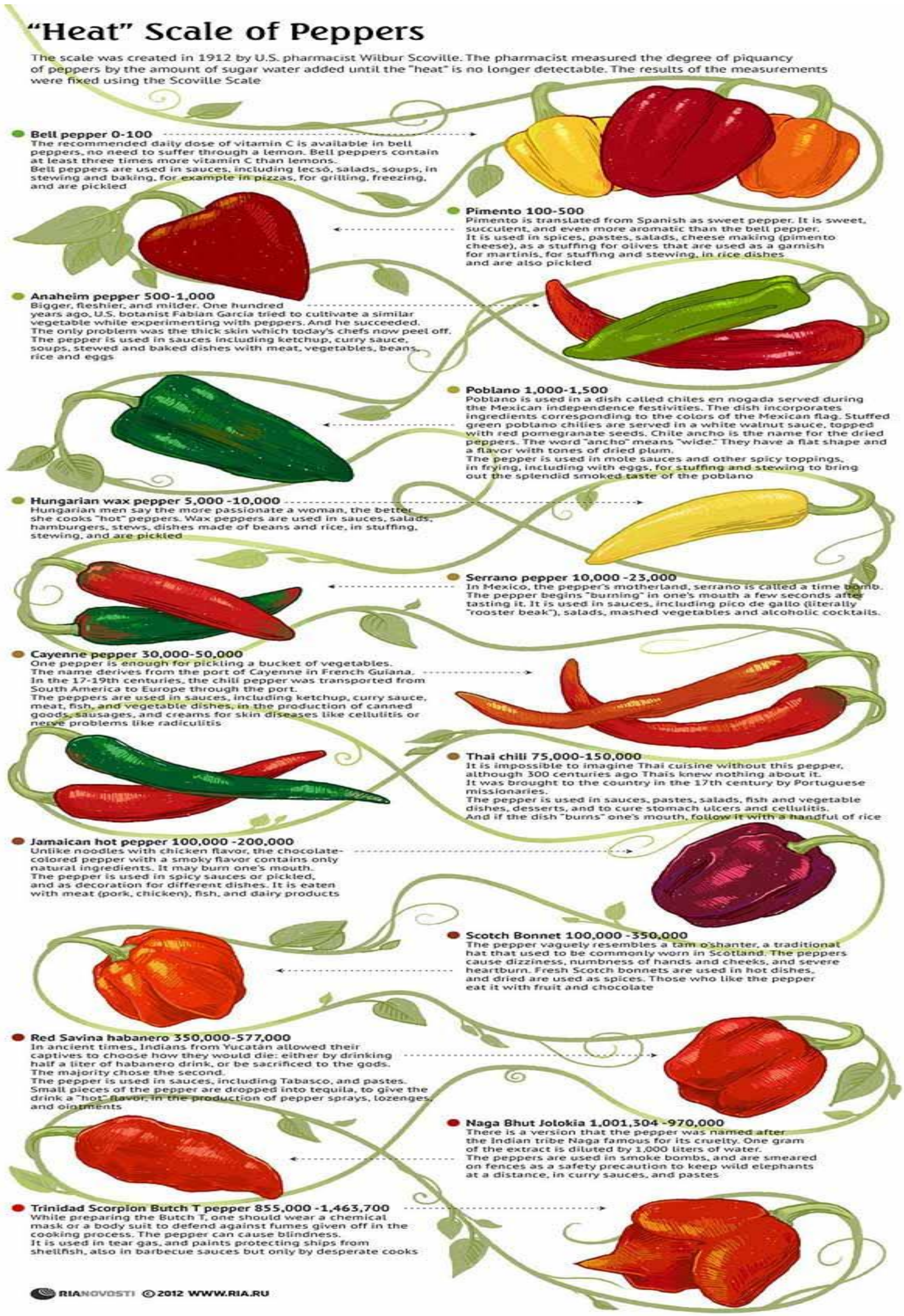
Proposed product

Pepper is considered a common spice used in a variety of cuisines. It's also used in cooking hot chips and chili dishes; moreover it is also main element in a variety of hot sauces, particularly those which are used as preservatives to vinegar.

All varieties of peppers belong (there are about 2500 species of chili and sweet) to Solanaceae fruits and contain high amounts of vitamin C, as a single fruit by weight of 75 grams contain the needs of an adult for one day of this important vitamin. They also contain good amounts of vitamin A, and the hot taste of pepper is attributed to the Capsicum. These types of pepper are used as food, whether it's fresh or cooked, and all turn to red when they reach full ripening.

Some surveys indicate that there are external demand on full ripen chili and of different Varieties (red) in a several European markets during the period from January to June and by relatively high prices compared to ordinary pepper prices and sweet pepper prices. And the project will produce the pepper of (Habanero) with a high quality specification (shape, color, taste, nutritional value). According to the Heat indicator of Scoville units of 100,000 – 350,000, often main ingredients of hot sauces are bottled. The color of Habaneros is green when it's unripe and when itripen their color will change, and the common colors are orange and red, but there is Brown and white and pink, it is most commonly used in Mexican kitchen, so one should be careful when dealing with it in the field and while preparation. The following illustrations show varieties of pepper habanero type and severity of temperature on the scale "skovli" severity hot pepper.

http://www.fotosearch.com/photos-images/habanero.html


"Heat" Scale of Peppers
The scale was created in 1912 by U.S. pharmacist Wilbur Scoville. The pharmacist measured the degree of piquancy of peppers by the amount of sugar water added until the "heat" is no longer detectable. The results of the measurements were fixed using the Scoville Scale
Bell pepper 0-100
The recommended daily dose of vitamin C is available in bell peppers, no need to suffer through a lemon. Bell peppers contain at least three times more vitamin C than lemons. Bell peppers are used in sauces, including lecsó, salads, soups, in stewing and baking, for example in pizzas, for grilling, freezing, and are pickled
Pimento 100-500
Pimento is translated from Spanish as sweet pepper. It is sweet, succulent, and even more aromatic than the bell pepper. It is used in spices, pastes, salads, cheese making (pimento cheese), as a stuffing for olives that are used as a garnish for martinis, for stuffing and stewing, in rice dishes and are also pickled
Anaheim pepper 500-1,000
Bigger, fleshier, and milder. One hundred years ago, U.S. botanist Fabian Garcia tried to cultivate a similar vegetable while experimenting with peppers. And he succeeded. The only problem was the thick skin which today's chefs now peel off. The pepper is used in sauces including ketchup, curry sauce, soups, stewed and baked dishes with meat, vegetables, beans, rice and eggs
Poblano 1,000-1,500
Poblano is used in a dish called chiles en nogada served during the Mexican independence festivities. The dish incorporates ingredients corresponding to the colors of the Mexican flag. Stuffed green poblano chilies are served in a white walnut sauce, topped with red pomegranate seeds. Chile ancho is the name for the dried peppers. The word "ancho" means "wide." They have a flat shape and a flavor with tones of dried plum. The pepper is used in mole sauces and other spicy toppings, in frying, including with eggs, for stuffing and stewing to bring out the splendid smoked taste of the poblano
Hungarian wax pepper 5,000-10,000
Hungarian men say the more passionate a woman, the better she cooks "hot" peppers. Wax peppers are used in sauces, salads, hamburgers, stews, dishes made of beans and rice, in stuffing, stewing, and are pickled
Serrano pepper 10,000-23,000
In Mexico, the pepper's motherland, serrano is called a time bomb. The pepper begins "burning" in one's mouth a few seconds after tasting it. It is used in sauces, including pico de gallo (literally "rooster beak"), salads, mashed vegetables and alcoholic cocktails.
Cayenne pepper 30,000-50,000
One pepper is enough for pickling a bucket of vegetables. The name derives from the port of Cayenne in French Guiana. In the 17-19th centuries, the chili pepper was transported from South America to Europe through the port. The peppers are used in sauces, including ketchup, curry sauce, meat, fish, and vegetable dishes, in the production of canned goods, sausages, and creams for skin diseases like cellulitis or nerve problems like radiculitis
Thai chili 75,000-150,000
It is impossible to imagine Thai cuisine without this pepper, although 300 centuries ago Thais knew nothing about it. It was brought to the country in the 17th century by Portuguese missionaries. The pepper is used in sauces, pastes, salads, fish and vegetable dishes, desserts, and to cure stomach ulcers and cellulitis. And if the dish "burns" one's mouth, follow it with a handful of rice
Jamaican hot pepper 100,000-200,000
Unlike noodles with chicken flavor, the chocolate-colored pepper with a smoky flavor contains only natural ingredients. It may burn one's mouth. The pepper is used in spicy sauces or pickled, and as decoration for different dishes. It is eaten with meat (pork, chicken), fish, and dairy products
Scotch Bonnet 100,000-350,000
The pepper vaguely resembles a tam o'shanter, a traditional hat that used to be commonly worn in Scotland. The peppers cause dizziness, numbness of hands and cheeks, and severe heartburn. Fresh Scotch bonnets are used in hot dishes, and dried are used as spices. Those who like the pepper eat it with fruit and chocolate
Red Savina habanero 350,000-577,000
In ancient times, Indians from Yucatán allowed their captives to choose how they would die: either by drinking half a liter of habanero drink, or be sacrificed to the gods. The majority chose the second. The pepper is used in sauces, including Tabasco, and pastes. Small pieces of the pepper are dropped into tequila, to give the drink a "hot" flavor, in the production of pepper sprays, lozenges, and ointments
Naga Bhut Jolokia 1,001,304-970,000
There is a version that the pepper was named after the Indian tribe Naga famous for its cruelty. One gram of the extract is diluted by 1,000 liters of water. The peppers are used in smoke bombs, and are smeared on fences as a safety precaution to keep wild elephants at a distance, in curry sauces, and pastes
Trinidad Scorpion Butch T pepper 855,000-1,463,700
While preparing the Butch T, one should wear a chemical mask or a body suit to defend against fumes given off in the cooking process. The pepper can cause blindness. It is used in tear gas, and paints protecting ships from shellfish, also in barbecue sauces but only by desperate cooks
RIANOVOSTI © 2012 WWW.RIA.RU

Scoville heat units	Name of peppers
855,000 – 2,200,000	Komodo Dragon Chili Pepper,[13] Trinidad Moruga Scorpion,[14] Naga Viper pepper,[15][16] Infinity Chilli,[17] Naga Morich, Bhut jolokia (ghost pepper),[18][19] Trinidad Scorpion Butch T pepper,[20] Bedfordshire Super Naga,[21] Spanish Naga Chili,[22] Carolina Reaper
350,000 – 580,000	Red Savina habanero[16][23][24]
100,000 – 350,000	Habanero chili,[3][25] Scotch bonnet pepper,[25] Datil pepper, Rocoto, Madame Jeanette, Peruvian White Habanero,[26] Jamaican hot pepper,[25] Fatalii[27] Wiri Wiri,[28] Bird's eye chili[29]
50,000 – 100,000	Malagueta pepper,[29] Chiltepin pepper, Piri piri, Pequin pepper,[29] Siling Labuyo, Capsicum Apache
30,000 – 50,000	Guntur chilli, Cayenne pepper, Ají pepper,[29] Tabasco pepper, Capsicum chinense
10,000 – 30,000	Byadgi chilli, Serrano pepper, Peter pepper, Chile de árbol, Aleppo pepper, Chungyang Red Pepper, Peperoncino
3,500 – 10,000	Guajillo pepper, 'Fresno Chili' pepper, Jalapeño, wax[30] (e.g. Hungarian wax pepper)
1,000 – 3,500	Gochujang, Pasilla pepper, Peppadew, poblano (or ancho),[31] Poblano verde,[30] Rocotillo pepper, Espelette pepper
100 – 1,000	Banana pepper, Cubanelle, paprika,[31] Pimento
0	Bell pepper[32][33]

Market size analysis

The demand for pepper in increasing both externally and internally to be used in various types of food, particularly of Mexican and Indian meals, this encourages producers to increase production and attract new producers to enter the market.

Increased demand is returning to the following reasons:

1. Change in consumption pattern in most peoples as well as the Jordanian consumer
2. Development of the tourism sector and the increasing number of hotels and restaurants.
3. Permanent increase in demand in the external market.

Department of statistics figures indicate that production of chili had reached 13 000 ton by 2015 (table 3)

Table 2 Growth of the cultivated area and the production of chili in Jordan

Year	Wintry			Summery			Total		
	Area	Production rate	Production	Area	Production rate	Production	Area	Production rate	production
2011	3,076	3.22	9,900	2,563	2.32	5,950	5,639	2.81	15,850
2012	1,923	3.88	7,464	3,496	2.37	8,294	5,419	2.91	15,758
2013	3,162	3.33	10,523	3,868	2.33	9,005	7,030	2.78	19,528
2014	2,627	0.58	1,512	2,843	0.56	1,587	5,470	0.57	3,098
2015	2,673	1.02	2,717	7,598	1.36	10,361	10,271	1.27	13,078

Source: Department of statistics

As for the import of chilli, no accurate statistics are available about the hot pepper, where it is merged with sweet pepper in the same items of the foreign trade, importers reported that the local market need to import various kinds of pepper is confined between 15/11-30/11 where this period being a transitional so that when Sheifa'a season ends and production in Alaghwar didn't start yet , the imported quantities for the last three years ranged between 115 – 490 tons as shown in the following table (table 4)

Table3 Evolution of the exports and imports of chili and sweet in Jordan

Year	Value of imports (Thous.)	Amount of importes (ton)	Value of national exports (Thous.)	Amount of national exports (ton)	Value of re-exports (Thous.)	Amount of re-exports (ton)	Value of total exports (Thous.)	Balance of trade (exports-:(imports (Thous.)
2011	166	230	23,675	32,233	5	6	23,680	23,514
2012	260	330	25,463	28,837	9	10	25,473	25,212
2013	489	722	25,952	35,278	0	0	25,952	25,463
2014	115	181	32,410	36,639	0	0	32,410	32,295
2015	166	172	39,741	51,522	0	0	39,741	39,575

As no data is available and accurate information about the size of actual imports or exports of chilies, EU data of exports of chilies to the EU countries had been used as an indicator of exports of chili for the European market. When reviewing the European Union data of imports from Jordan of Chili, it show that there's annual exports ranging from (1,787) tons. (Table 5).

Table 5 shows the quantities of import, export and domestic production and overall demand for chili during the period (2010-2015), and Jordan's exports of chilies for the year 2012 was about 3,228 ton, mostly to Hungary and Romania. And the highest prices of exported chili was to Netherlands markets and United Kingdom markets at prices ranging from (3-4) euros per a kg (table 6). Chili decline of export to EU countries in recent years is attributed to the investor who was cultivating chili and exporting it to European markets and abandoned this field and turned it into a more feasable projects to produce seeds of chili and sweet pepper and other vegetables.To comparison between chili and sweet pepper export, table (7) shown exports quantity of sweet pepper to the EU average of 3,700 ton annually.

Table 4 EU imports of chilli from Jordan by country in tons

Country	2010	2011	2012	2013	2014	2015
Austria	0	16	0	16	9	0
Belgium	2	0	0	0	0	0
Bulgaria	9	3	4	0	0	0
Croatia	0	0	0	0	9	0
Denmark	2	4	0	0	1	0
France	0	0	0	0	0	0
Germany	10	11	13	11	14	12
Greece	17	15	0	0	0	0
Hungary	1,049	877	580	751	417	16
Ireland	0	0	0	0	0	0
Netherlands	19	6	5	5	1	0
Poland	0	0	0	0	0	0
Romania	21	1,318	2,437	1,540	115	22
Slovenia	0	26	0	11	287	432
Spain	0	3	6	0	0	0
Sweden	0	0	0	0	0	0
United Kingdom	102	109	183	116	51	39
European Union countries	1,231	2,388	3,228	2,450	904	521

source:http://exporthelp.europa.eu/thdapp/display.htm?page=st%2fst_Statistics.html&docType=main&languageId=en

Table 5 Average price of 1 kilogram of EU imports of chilli from Jordan State

Country	Export price (euro/kg)					
	2010	2011	2012	2013	2014	2015
Austria		1.945		1.774	0.899	
Belgium	0.157					
Bulgaria	0.277	0.403	0.355			
Croatia					1.364	
Denmark	0.590	0.618			1.346	
France						

Country	Export price (euro/kg)					
	2010	2011	2012	2013	2014	2015
Germany	0.431	0.424	0.373	0.293	0.298	0.497
Greece	1.898	1.471				
Hungary	1.374	1.643	1.511	1.587	1.652	0.884
Ireland						
Netherlands	**2.322**	**2.783**	**2.357**	**1.206**	**2.222**	
Poland						
Romania	0.189	0.151	0.264	0.443	0.888	1.747
Slovenia		1.716		1.337	1.259	1.113
Spain		3.854	2.746			
Sweden						
United Kingdom	**2.994**	**2.891**	**3.850**	**3.795**	**3.978**	**4.717**
European Union countries	**1.494**	**0.876**	**0.701**	**0.969**	**1.535**	**1.394**

Source: http://exporthelp.europa.eu/thdapp/display.htm?page=st%2fst_Statistics.html&docType=main&languageId=en

Table 6 The European Union imports of sweet peeper from Jordan by country

السنة	Amount of exports (ton)				
	2011	2012	2013	2014	2015
Austria	53	0	21	420	0
Belgium	124	137	12	0	0
Croatia	88	65	39	32	0
Czech Republic	14	14	0	0	0
Denmark	139	0	0	0	0
Germany	0	0	0	0	0
Hungary	1,750	2,697	2,680	524	19
Netherlands	0	0	176	0	0
Poland	18	0	43	0	0
Romania	3,746	2,111	970	615	365
Slovenia	75	120	37	919	903
Sweden	0	0	0	0	0
United Kingdom	0	5	0	1	0
European Union countries	**6,007**	**5,149**	**3,978**	**2,511**	**1,287**

Due to the difficulty of separating the foreign trade data between chili and sweet peppers, imports and exports of various kinds of peppers are shown in table (8), where the demand of hot peppers was estimated in two ways that lead to the same result, the first method based on the general trend of domestic production of Chili, the second method of calculating per capita consumption rate where the quantities of demand (domestic production + net imports-exports) wee divided on the number of population in previous years, to obtain the share per capita that amounts to (3.5) kg per capita , and as the consumption rate is related to the population as a result of increasing demand for food, and where the rate of population growth in the Kingdom according to the Department of statistics is 2.2% per year, assuming that the demand growth rate equal to the rate of population growth. Domestic demand is expected to chili during the period (2018 – 2027) as shown in the table (9)

.

Table 7 Evolution of imports and domestic production of different kinds of pepper during the period (2011-2017) in tons

Jordan market size of chili	2011	2012	2013	2014	2015	2016	2017*
Imports (ton)	230	330	722	181	172	176	179
Exports (ton)	32,233	28,837	35,278	36,639	51,522	53,068	54,660
Re-exports (ton)	6	10	0	0	0	0	0
Domestic production of goods (ton)	63,738	72,178	60,868	66,103	86,677	88,410	90,179
Overall demand for product (ton)	31,269	43,000	24,868	29,283	34,982	35,682	36,396

Source: Department of statistics (2015), (2) CBE (2015)

Table 8 Estimate the expected demand for different kinds of peppers (hot and sweet) (2018 – 2027) in ton

Amount of the demand of product	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
The expected population (thous. person)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual rate per capita consumption (kg/capita)	3.500	3.500	3.500	3.500	3.500	3.500	3.500	3.500	3.500	3.500
Projected demand of chili (ton)	34,945	35,714	36,499	37,302	38,123	38,962	39,819	40,695	41,590	42,505
Expected production of project/chili (ton)	14	27	27	27	27	27	27	27	27	27
Market share of project of domestic demand (%)	0.0%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%

Source: team of study

Compititors analysis

The project is supposed to market most of its production to export markets especially to European markets in general and Dutch and British in particular due to the large presence of Indian origins communities and the Mexican and Latin American origins, where these varieties enter in traditional dishes of the citizens of these countries. The production of ripe chilies (red) is well received and well priced. This data has been shown by Dr. Fouad Salamah: that he was able to export reasonable quantities during the last years for the two above mentioned markets (Dutch and British) with prices ranging from € 3 to € 6 per kilogram. This project is not expected to face difficulties in marketing its products whether in the domestic market or foreign markets if the production was according to the specific production specifications in the market.

For the purposes of this study, it was considered that the selling price per ton 2000 JD of first class by a ratio (40%) and 1500 dinars per ton of second class by a ratio (30%) and the dried by a ratio (30%), and the percentage of dry material of it (20%) for sold after the manufacturing 15 dinars in foreign markets and the local market. The ratio of exports to total production constituted 80%. It is reach the annual project revenues to about 70 thousand dinars.

Prices analysis

Jordan exports of chili since 2012 amounted to about 3,228 ton, mostly of it was to Hungary and Romania. And thehighest prices of chili that was exported to the Netherlands market and United Kingdom market at prices ranging between (3-4) euros per kg.So it is suggested that the price of one kilogram of chili (2) JD per kg of first class and price (1.5) Jd per kg of the second class and dry of manufactured in plastic bottles by (15) Jd per kg.

Market share

The rate of production projected of donom is between 5-8 ton/donom, and for the purposes of this study assumed the production of one donom by 3.6 ton, where the external market share of it 80% and local dried by 20% in the first year and by the growth rate 2.2% for the coming years, and thus the production volume be 18 ton per year and annual production 36 ton, where the it was planted of a land area to be 5 donom of 10 donom, the project share is about 1% of total local production, its a modest share and this confirms the possibility of marketed locally and externaly and that the project will not face any marketing problem.

This survey has shown that coloured sweet pepper has become required and it has consumers at most times, especially those with low production volume. Selling price per ton of regular chili between 1500-2000 dinars in the local market and doubled of it in the external market, to study purposes we assume price of ton 2,850 dinars per ton as a weighted average of different grades.

Projected revenues

The project covers cultivation 10 of plastic houses at the rate of 500 meter square per house, as the productive per donom for this type of cultivation between 3-4 ton, for the purposes of this study, it was considered that the rate of productivity per donom is 3.5 ton, where 80% of which will be sold in export markets, so the project will produce about 21 ton sold of which for export markets about 14 ton and will be converted to pepper dried to sold in the domestic market and export, for the purposes of this study, was considered that the weighted average of selling price per ton (2,828) JD/ton of chili, are expected to increase sales and purchase prices for chili by 2% annually.

Table 9 Sale prices and expected sales total during the period (2018-2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Chilli prices (JD/ton) firs class	2.0	2.0	2.1	2.1	2.2	2.2	2.3	2.3	2.3	2.4
Chilli prices (JD/ton) second class	1.5	1.5	1.6	1.6	1.6	1.7	1.7	1.7	1.8	1.8
Chilli prices (JD/ton) dry class	15.0	15.3	15.6	15.9	16.2	16.6	16.9	17.2	17.6	17.9
Amount of first class	11,100	22,644	23,097	23,559	24,030	24,511	25,001	25,501	26,011	26,531
Amount of second class	8,325	16,983	17,323	17,669	18,022	18,383	18,751	19,126	19,508	19,898
Amount of dry class	22,200	45,288	46,194	47,118	48,060	49,021	50,002	51,002	52,022	53,062
Total sales (Thous. of JD)	**41,625**	**84,915**	**86,613**	**88,346**	**90,112**	**91,915**	**93,753**	**95,628**	**97,541**	**99,491**

3. Technical analysis

One dunum needs about 60 g of seeds by rate1600 seedlings for one plastic house and organic fertilizer at a rate of 1 ton, and nitrogen fertilizers, superphosphate fertilizer. This is in addition to pesticides, water and land preparation for cultivation. The following table shows the the project's needs of those costs

Required raw material for production

For the production of hot pepper there are many suppliments are needed for the production, andfor one dunum about 60 g of seeds by rate 1600 seedlings is needed for one plastic house and organic fertilizer at a rate of 1 ton, and nitrogen fertilizers, superphosphate fertilizer. This is in addition to pesticides, water and land preparation for cultivation. The following table shows the requirements of the project of those costs

Table 10 Materials required for one plastic house of production suppliments

Variable costs	Unit	Amount	Price/unit	Value
Production requirements				
seeds	Seedling/House	1500	0.05	75.0
Chemical fertilizer compound	Kg/house	50	0.6	30.0
Rare elements fertilizer	L/house	1	8.5	8.5
Organic fertilizer	Ton/house	1	20	20.0
Herbicides	L/house	1	10	10.0
Insecticides	L/house	1	32	32.0
Soil sterilization	Section/house	1	25	25.0
Mulch black	Kg	6	2	12.0
others		1	20	20.0
cans	can	500	0.4	200.0
water	m^3	500	0.2	100.0
Total				532.5

Human resources required

Chilli production needs five unskilled workers, working in one shift a day, in addition to agricultural engineer specialized in plant production has a whole experience in chili production. One person is enough to manage this farm, assuming that the project manager will dedicate his time to making deals with the exporters and product marketing and to negotiate the purchase of production requirments of crop residues and animal dung, and his possibly might need to a second person (production supervisor) for field supervision of production process. The following table shows the employment requirements.

Table 11 manpower required

Job title	No.
Indirect workers	
Project Manager/plant production engineer	1
Production supervisor	1
Total	**2**
Direct workers	
Workers of cultivation	3
Manufacturing workers	2
Total	**5**
Total Sum.	**7**

- **Agricultural Engineer**: A well knowledge of how to produce, manufacture chili and how to treat it, the ability to request and provide the necessary materials of tools and requirements of production, such as seeds and seedlings, fertilizers, pesticides required. Implementation of preventive measures and control of agricultural pests in the product in accordance with environmental standards and requirements of integrated pest Management control. Follow-up accreditation to meet the requirements of Global GAP. Preparation and packaging, as well as the mobilization of farm records and business models, documenting expenditures, proposing policies and action plans, monitoring implementation and evaluating the results. Organizing daily work, distributing workers among tasks and following up the implementation of their work in coordination with farmers, suppliers and transport companies to ensure the safety of raw materials supply and production.

- **Production Supervisor**: Soil preparation for cultivation such as tillage, plowing, leveling, installing drop irrigation, and field planning. Securing the materials and supplies for propagation of seeds, fertilizers and pesticides. Cultivation of seeds of different varieties according to plans. Adding chemical and organic fertilizers, operation of irrigation systems according to prepared programs, implementation of prevention and control programs according to guidelines, soil weeding and prevent erosion, removal of unrepresented plants, monitoring stages of growth, development, flowers and maturity. assist in hybridization of species according to the guidelines and accordance to environmental standards. Harvesting of crops, drying, extraction of seeds, sifting and sorting of seeds, treatment and disinfection, packaging and preservation. Filling farm records and business models. Evaluate the performance of staff and labors, and to identify their training needs, and training and skills development. Applying occupational safety and health procedures and instructions
- **Agricultural worker**: Works under the supervision of the agricultural engineer and prepares the necessary materials for the process of agriculture, care, fertilizers, irrigation, irrigation scheduling, harvesting, processing, packaging and the application of occupational safety and health procedures and regulations.
- **Manufacturing worker**: Preparation of packaging tools and containers and, scissors and knives. Receiving the crop and unloading it in the designated area. Place the field containers on the production line, and sorting the fruits distorted and infected products from healthy fruits, classification of fruits according to the required specifications. Fill the fruit in the custom packaging and pack it according to instructions. Collection and packaging of containers, and storage in warehouses or refrigerators until marketing. Applying occupational safety and health procedures and instructions.

Side analysis

Pepper in general needs a mild climate that slightly inclined to heat, but it does not bear coldness. Plants do not grow at temperatures below 6 Celsiusand do not bear frost, while it resists high temperatures and drought more than the rest of the family Solanaceae as tomatoes and potatoes.Thus the degree of (21-27) is considered the best for the growth and production, and irregular irrigation operations and inappropriate temperatures lead to the fall of the buds and flowers.

The clay sandy soil is rich in organic materials is considered the most appropriate land and soil to be well drained, as the rate of acidity in it ranges between (5-7) degrees. But Peppers is sensitive to high land of gigh level of sulphur dioxide. So, Alaghwar and Sheifa'a are the most suitable for this type of cultivation. Jerash is suitable for producing chili for the following reasons:

1. Provide adequate water quantity and quality.
2. The mild climate which greatly reduces of risk factors (snow, Frost, high heat).
3. Average location near services.
4. Length of production Period.

According to all these data, it is proposed to establish in the Jarash a project for the production of chili varieties, so that the the appropriate location for production is chosen based on the availability of infrastructure services of water and electricity and the availability of raw materials,

Description of production process and manufacturing steps

The project covers the cultivation of 10 green houses at the rate of 500 meter square per house, as the production per dunum for this type of cultivation ranges between 3-4 ton. For the purposes of this study, it was considered that the rate of productivity per dunum is 3.5 ton, where 70% of which will be sold in export markets, that's the project will produce about 21 ton sold of which 14 tons will be for export markets and will be converted to dried pepper dried to be sold in the domestic market and export.

1. Choose, surveying and flattening side.
2. Project planning and identification and drilling cement bases.
3. Installation of the metal structure.
4. Installation of nets and installation of plastic.
5. Installation of of irrigation net and fertilisation and filtration unit.
6. Land working (plowing, terracing, put plastic, soil, planting seedlings).

Technical requirements analysis

Cultivation and production process and drying chili peppers are summarized in the following steps:

Land preparation for cultivation

Installation of green houses and supplies

Cultivation Pepper seedlings of required exportable items.

Caring for the crop of fertilization, irrigation scheduling and agricultural pest control

Picking, receiving, sorting crop and cleaning them

Packaging of Varieties by grading and marketing degrees required.

To manufacture other grades the following processes are needed:

- Washing products with water
- Cutting products received and preparation for drying
- Place the products in wooden container and expose them to the Sun for drying.
- Grind the dried products by using special grinding machines
- Filling the finished product in special plastic container and packaging them.

Land and construction works

The p roject needs about 10-donum , the green house area approximatelt 500 m², the production area will be (10 green houses) about 5 dunum, however, 10 dunum will be rented for the purpose of providing services for future expansion, and the achievement of Global GAP it requires independent buildings for management and workers ' housing and separate warehouses for pesticides and fertilizers, refrigerate warehouses for the products, a separate place to arrange the productsseparate showers and baths for workers , the rest of space for the movement and processing of materials and the remaining space is devoted to services, facilities and corridors.

Table 12 land required for the project

Land	Area (m2)
Land required	**10,000**

Regarding the construction required for the project are as follows:

Table 13 Construction required

Construction works	Area (m2)
Administrative offices	50

Storage warehouse for pesticides	20
Warehouse storage fertilizers	20
Refrigerated warehouse	50
Room and their accessories for workers	50
Assembly and packaging department	60
baths for workers	30

Machinery and equipment

The equipment is consists mainly of plastic houses and their supplies as well as supplies for the drying process, grinding and packaging. The following table shows the machinery and equipment needed for the project and quantity as follows:

Table 14 Machines and equipment

Machines and equipment	No.
Plastic houses with installation	10
Gauze, plastic covers	20
Earthen pond covered with plastic.	1
motor pumping water (diesel)	1
Drip irrigation system	10
Motor spray	1
Water filtration unit (sand filter + filter networks)	1
Fertilizing unit	1
Delicate balance	2
Packaging unit	1
Chopping machine for pepper	1
Milling machine 20 kg / hour.	1
Semi automatic packaging machine	1
Work tables and hand tools for production	4
Hot-air drying oven with conveyor belt	1
Wooden drying trays	50
Different tools	1

Furnishings

Needs of project of furniture and furnishings:

Table 15 Furniture required for the project

furniture and furnishings	No.
Office furniture	1
Computer	1
Kitchen and cafeteria	1

Vehicles and transportation

The following table shows the vehicles and transportation required for the project:

Table 16 Vehicles and transportation

Vehicles	No.	Unit cost/JOD
Medium transport vehicle	1	15,000

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table : 3 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table : 4 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table : 5 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table : 6 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	1.00%
Maintenance Expenses from Total Revenues	0.05%
Depreciation Expenses from Total Revenues	0.05%
Insurance Expenses from Total Assets Value	0.00%
Marketing Expenses from Total Revenues	2.00%

Table : 7 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	100%
Income Tax after Deduction	0%
(3) Compulsary Reserve Percentage	10%
(1) Average Income Tax in Jordan	القيمة
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table : 8 Risk premium assumptions

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premuim	9.93%
Income Tax Rate	0.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table : 9 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	52,901
Loans Value	52,901

Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	52,901
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	52,901
Equety Percentage	%50
Gross	
Project Value	105,803
Weighted Average Cost of Capital	**%11.97**

Capital Expenditures

The estimated cost of the project (96,184) JD and it covers machinery and equipment costs (30,650) JD, construction works (27,820) JD, vehical and transportation cost (15,000) JD and working capital cost (16,714) JD. Where the working capital needed to buy production requirements and salaries, and will add 10% contingency reserve price as shown in the following tables

Table 10 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land		10,000	The rent interference in operational costs

Table 11 Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross
Administrative offices	120	50	6,000
Storage warehouse for pesticides	80	20	1,600
Warehouse storage fertilizers	80	20	1,600
Refrigerated warehouse	100	50	5,000
Room and their accessories for workers	80	50	4,000
Assembly and packaging department	80	60	4,800
baths for workers	100	30	3,000
Other works	-	-	1,820
Total		**280**	**27,820**

Table 12 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Plastic houses with installation	10	1,500	15,000
Gauze, plastic covers	20	200	4,000
Earthen pond covered with plastic.	1	1,000	1,000
motor pumping water (diesel)	1	500	500
Drip irrigation system	10	150	1,500
Motor spray	1	500	500
Water filtration unit (sand filter + filter	1	600	600
Fertilizing unit	1	500	500
Delicate balance	2	200	400
Packaging unit	1	100	100
Chopping machine for pepper	1	600	600
grinding machine 20kg an hour capacity	1	1,000	1,000
Semi automatic packaging machine	1	1,000	1,000
Work tables and hand tools for production	4	300	1,200
Hot-air drying oven with conveyor belt	1	2,000	2,000
Wooden drying trays	50	5	250
Different tools	1	500	500
Total			**30,650**

Table 13 Furniture and furnishings Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Office furniture	1	1,000	1,000
Computer	1	1,000	1,000
Kitchen and cafeteria	1	1,000	1,000
Total			3,000

Table 14 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	1	15,000	15,000
Total	**1**		**15,000**

Table 15 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	0
Marketing	2,000
Training	500
Total	**3,000**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 16 Working Capital

Working Capital	No. of Months	%	Estimated Cost
Operating expenses	6	%50	11,674
General and administrative expenses	3	%25	124
Marketing expenses	3	%25	117
Indirect Salaries	3	%25	4,800
Total			**16,714**

Table 17 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	0	10%	0	%0.0
Construction works	27,820	10%	30,602	%28.9
Machinery &Equipment	30,650	10%	33,715	%31.9
Furniture	3,000	10%	3,300	%3.1
Vehicles	15,000	10%	16,500	%15.6
Pre-Operating Expenses	3,000	10%	3,300	%3.1
Working capital	16,714	10%	18,386	%17.4
Total (JD)	**96,184**		**105,803**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (105,803) JD, where 50% of the project will be financed through loans (52,901) JD, and the remaining (50%) through (Equity) with (52,901) JD. The following table summarizes the general structure of the required financing:

Table 18 Sources of Funding

Funding Sources	Amount	Percentage
Equity	52,901	%50
Loans	52,901	%50
Total	**105,803**	**%100**
Use of Fund:	**Value**	**Ratio**
Capital Expenditures	84,117	%79.50
Pre-Operating Expenses	3,300	%3.12
Working Capital	18,386	%17.38
Total	**105,803**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 19 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cost of raw materials and water.	2,663	5,325	5,432	5,540	5,651	5,764	5,879	5,997	6,117	6,239
Rent land	500	500	510	520	531	541	552	563	574	586
Utilities from Revenues	19	39	40	41	42	43	44	44	45	46
Maintenance from Revenues	19	39	40	41	42	43	44	44	45	46
Disposables from Revenues	19	39	40	41	42	43	44	44	45	46
Direct Salaries	18,000	24,720	25,214	25,719	26,233	26,758	27,293	27,839	28,396	28,963
Others	2,122	3,066	3,128	3,190	3,254	3,319	3,385	3,453	3,522	3,593
Total	**23,343**	**33,730**	**34,404**	**35,092**	**35,794**	**36,510**	**37,240**	**37,985**	**38,745**	**39,520**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 20 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	200	204	208	212	216	221	225	230	234	239
Hospitality	100	102	104	106	108	110	113	115	117	120
Staionary and Printing	50	51	52	53	54	55	56	57	59	60
Transportaton	100	102	104	106	108	110	113	115	117	120
Training	0	0	0	0	0	0	0	0	0	0
Insurance	0	0	0	0	0	0	0	0	0	0
Miscellaneous	45	46	47	48	49	50	51	52	53	54
Total	**495**	**505**	**515**	**525**	**536**	**547**	**557**	**569**	**580**	**592**

Marketing Expenses

Table 21 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**467**	**675**	**688**	**702**	**716**	**730**	**745**	**760**	**775**	**791**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 22 The projected staff numbers

Job title	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Indirect										
Project Manager/plant production engineer	1	1	1	1	1	1	1	1	1	1
Production supervisor	1	1	1	1	1	1	1	1	1	1
Indirect employes	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
Direct										
Workers of cultivation	عمال	3	3	3	3	3	3	3	3	3
Manufacturing workers	عمال	2	2	2	2	2	2	2	2	2
Direct employes	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**
Total	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**

Table 23 Expected monthly salary and wages

Job Title	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Indirect										
Project Manager/plant production engineer	700	721	743	765	788	811	836	861	887	913
Production supervisor	500	515	530	546	563	580	597	615	633	652
Direct										
Workers of cultivation	300	309	318	328	338	348	358	369	380	391
Manufacturing workers	300	309	318	328	338	348	358	369	380	391

Table 24 Expected Annual Salaries and Wges

Job Title	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Project Manager/plant production engineer	11,200	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Production supervisor	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Total of indirect employes	**19,200**	**19,776**	**20,369**	**20,980**	**21,610**	**22,258**	**22,926**	**23,614**	**24,322**	**25,052**
Workers of cultivation	10,800	14,832	15,277	15,735	16,207	16,694	17,194	17,710	18,241	18,789
Manufacturing workers	7,200	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Total of direct employes	**18,000**	**24,720**	**25,462**	**26,225**	**27,012**	**27,823**	**28,657**	**29,517**	**30,402**	**31,315**
Total	**37,200**	**44,496**	**45,831**	**47,206**	**48,622**	**50,081**	**51,583**	**53,131**	**54,724**	**56,366**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 25 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	30,602	%5.0	%0.0
Machinaries	33,715	%10.0	%2.0
Furniture	3,300	%10.0	%5.0
Vehicles and trasportaions	16,500	%10.0	%0.0

Table 26 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2022	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	84,117	84,956	85,817	86,701	87,607	88,538	89,493	90,473	91,480	92,514
Total Depreciation Expenses	6,882	6,966	7,052	7,140	7,231	7,324	7,419	7,517	7,618	7,721
Total Accumulated Depreciation	6,882	13,847	20,899	28,039	35,269	42,593	50,012	57,529	65,147	72,869
Total Additions	0	839	861	883	907	930	955	980	1,007	1,034
Total Net Book Values	77,235	71,109	64,919	58,662	52,338	45,945	39,481	32,944	26,333	19,645

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table : 27 Loan Details

Loan Value	193,362
Annual Interst Rate	9.50%
Loans Period	5
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2017				52,901
2018	13,777	5,026	8,752	44,150
2019	13,777	4,194	9,583	34,566
2020	13,777	3,284	10,494	24,073
2021	13,777	2,287	11,491	12,582
2022	13,777	1,195	12,582	0

Income Statement

Table 28 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	41,625	84,915	86,613	88,346	90,112	91,915	93,753	95,628	97,541	99,491
Gross Operating Revenues	41,625	84,915	86,613	88,346	90,112	91,915	93,753	95,628	97,541	99,491
Operating Expenses	(23,347)	(33,740)	(34,414)	(35,103)	(35,805)	(36,521)	(37,251)	(37,996)	(38,756)	(39,531)
Gross Operating Profit	18,278	51,175	52,199	53,243	54,308	55,394	56,502	57,632	58,784	59,960
Gross Profit Percentage	*%44*	*%60*	*%60*	*%60*	*%60*	*%60*	*%60*	*%60*	*%60*	*%60*
Salaries and Benefits (Indirect Staff)	(19,200)	(19,776)	(20,369)	(20,980)	(21,610)	(22,258)	(22,926)	(23,614)	(24,322)	(25,052)
General and Administraive Expenses	(495)	(505)	(515)	(525)	(536)	(547)	(557)	(569)	(580)	(592)
Markeing Expenses	(467)	(675)	(688)	(702)	(716)	(730)	(745)	(760)	(775)	(791)
Pre Operating Expenses	(3,300)									
Gross Indirect Expenses	(23,462)	(20,956)	(21,573)	(22,208)	(22,862)	(23,535)	(24,228)	(24,942)	(25,677)	(26,434)
Income before Interest, Depreciation and Tax	(5,184)	30,220	30,626	31,035	31,446	31,859	32,273	32,690	33,107	33,526
Fixed Assets Depreciations	(6,882)	(6,966)	(7,052)	(7,140)	(7,231)	(7,324)	(7,419)	(7,517)	(7,618)	(7,721)
Income before Tax and Interests	(12,066)	23,254	23,575	23,895	24,215	24,535	24,854	25,172	25,489	25,805
Bank Interests	(5,026)	(4,194)	(3,284)	(2,287)	(1,195)	0	0	0	0	0
Income Before Tax	(17,092)	19,060	20,291	21,608	23,020	24,535	24,854	25,172	25,489	25,805
Income Tax	0	0	0	0	0	0	0	0	0	0
Net Profit	(17,092)	19,060	20,291	21,608	23,020	24,535	24,854	25,172	25,489	25,805
Net Profit Percentage	*%41-*	*%22*	*%23*	*%24*	*%26*	*%27*	*%27*	*%26*	*%26*	*%26*

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Compulsory Reserves	0	(1,906)	(2,029)	(2,161)	(2,302)	(2,454)	(2,485)	(2,517)	(2,549)	(2,580)
Retained Earnings	(17,092)	62	18,324	37,772	58,490	80,571	102,940	125,596	148,536	171,761

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 29 Expected Cashflows

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Cash Inflows from Operating Activities										
Net Profit	(17,092)	19,060	20,291	21,608	23,020	24,535	24,854	25,172	25,489	25,805
Bank Interests	5,026	4,194	3,284	2,287	1,195	0	0	0	0	0
Depreciation	6,882	6,966	7,052	7,140	7,231	7,324	7,419	7,517	7,618	7,721
Total Operating Cashflows before Additions to Working Capital	(5,184)	30,220	30,626	31,035	31,446	31,859	32,273	32,690	33,107	33,526
Inventory (Increase/Decrease)	(222)	(222)	(9)	(9)	(9)	(9)	(10)	(10)	(10)	(10)
Accounts Receviables (Increase/Decrease)	(3,469)	(3,608)	(142)	(144)	(147)	(150)	(153)	(156)	(159)	(163)
Accounts Paybales (Increase/Decrease)	1,724	644	47	48	49	50	51	52	53	54
Working Capital (Increase/Decrease)	(1,967)	(3,185)	(103)	(105)	(107)	(109)	(112)	(114)	(116)	(118)
Net Cashflows from Operating Activities	(7,151)	27,034	30,523	30,930	31,339	31,750	32,162	32,576	32,991	33,408

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(84,117)	(839)	(861)	(883)	(907)	(930)	(955)	(980)	(1,007)	(1,034)
Net Cashflows from Investments Activities	(84,117)	(839)	(861)	(883)	(907)	(930)	(955)	(980)	(1,007)	(1,034)
Cashflows from Financing Activities										
Capital	52,901									
Loan Amortization	(8,752)	(9,583)	(10,494)	(11,491)	(12,582)	0	0	0	0	0
Bank Interest Rate	(5,026)	(4,194)	(3,284)	(2,287)	(1,195)	0	0	0	0	0
Loans	52,901	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	92,025	(13,777)	(13,777)	(13,777)	(13,777)	0	0	0	0	0
Net (Increase/Decrease) in Cash	757	12,418	15,885	16,269	16,655	30,819	31,207	31,595	31,985	32,374
Cashflows at the Beginning of Period	0	757	13,175	29,059	45,329	61,984	92,803	124,010	155,605	187,590
Cashflows at the End of Period	757	13,175	29,059	45,329	61,984	92,803	124,010	155,605	187,590	219,964

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 30 Expected Balance Sheet

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Assets										
Current Assets										
Cash	757	13,175	29,059	45,329	61,984	92,803	124,010	155,605	187,590	219,964
Inventory	222	444	453	462	471	480	490	500	510	520
Accounts Receivable	3,469	7,076	7,218	7,362	7,509	7,660	7,813	7,969	8,128	8,291
Total current Assets	**4,448**	**20,695**	**36,730**	**53,152**	**69,964**	**100,943**	**132,312**	**164,074**	**196,228**	**228,775**
Non Current Assets										
Fixed Assets (net)	77,235	71,109	64,919	58,662	52,338	45,945	39,481	32,944	26,333	19,645
Total Non Current Assets	**77,235**	**71,109**	**64,919**	**58,662**	**52,338**	**45,945**	**39,481**	**32,944**	**26,333**	**19,645**
Total Assets	**81,683**	**91,804**	**101,648**	**111,815**	**122,302**	**146,887**	**171,793**	**197,018**	**222,560**	**248,420**
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	1,724	2,368	2,415	2,464	2,513	2,563	2,614	2,667	2,720	2,774
Remaining amount of Loan	9,583	10,494	11,491	12,582	0	0	0	0	0	0

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Total current Liabilities	**11,307**	**12,862**	**13,906**	**15,046**	**2,513**	**2,563**	**2,614**	**2,667**	**2,720**	**2,774**
Non Current Liabilities										
Long Terms Loans	34,566	24,073	12,582	0	0	0	0	0	0	0
Total Long Term Liabilities	**34,566**	**24,073**	**12,582**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total Liabilities	**45,873**	**36,934**	**26,488**	**15,046**	**2,513**	**2,563**	**2,614**	**2,667**	**2,720**	**2,774**
Owners Equity										
Shareholders Contributions	52,901	52,901	52,901	52,901	52,901	52,901	52,901	52,901	52,901	52,901
Statutory Reseve	0	1,906	3,935	6,096	8,398	10,851	13,337	15,854	18,403	20,984
Retained Profits	(17,092)	62	18,324	37,772	58,490	80,571	102,940	125,596	148,536	171,761
Total Equity	35,810	54,870	75,161	96,769	119,789	144,324	169,178	194,351	219,840	245,645
Total Liabilities and Equity	**81,683**	**91,804**	**101,648**	**111,815**	**122,302**	**146,887**	**171,793**	**197,018**	**222,560**	**248,420**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Free Net Cash flows Table 31 Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(105,803)	(3,851)	26,195	29,662	30,047	30,432	30,819	31,207	31,595	31,985	32,374	422,713
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	**(105,803)**	**(3,440)**	**20,896**	**21,133**	**19,119**	**17,295**	**15,643**	**14,147**	**12,793**	**11,566**	**10,456**	**136,528**

Table below (49) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 32 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(105,803)	(3,851)	26,195	29,662	30,047	30,432	30,819	31,207	31,595	31,985	32,374
Project Value Returned	105,803	109,654	83,459	53,796	23,750	(6,682)	(37,502)	(68,708)	(100,304)	(132,288)	(164,663)
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table :33 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	**%11.97**
Net Present Value for Cashflows	**170,335**
Payback Period	**4**
Internal Rate of Return	**%26.58**

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table :34 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	Sensitivity Analysis
Original Scenario	26.58%	4	170	11.97%
Revenues declined by 10%	19.59%	6	82	11.97%
Operating Expenses Increased by 10%	23.48%	5	131	11.97%
Rising total costs 10%	21.84%	5	121	11.97%
High costs and low returns 10% together	14.53%	8	33	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 170,335 Jordan Dinars considering that the project provides 7 Job Opportunitites for the governorate residents.